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Non-Symmetrical Dielectric Relaxation Behaviour Arising from a Simple Empirical Decay Function

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Received 13*th June*, 1969

The empirical dielectric decay function $\gamma(t) = \exp -(t/\tau_0)^\beta$ may be transformed analytically to give the frequency dependent complex dielectric constant if β is chosen to be 0.50. The resulting dielectric constant and dielectric loss curves are non-symmetrical about the logarithm of the frequency of maximum loss, and are intermediate between the Cole-Cole and Davidson-Cole empirical relations. Using a short extrapolation procedure, good agreement is obtained between the empirical representation and the experimental curves for the α relaxation in polyethyl acrylate. It is suggested that the present representation would have a general application to the α relaxations in other polymers. The Hamon approximation, with a small applied correction, is valid for the present function with $\beta = 0.50$ in the range $\log(\omega\tau_0) > -0.5$, but cannot be used at lower frequencies.

The dielectric dispersions and absorptions of many systems do not correspond to a single relaxation time expression. Where the behaviour is symmetrical about the logarithm of the frequency of maximum loss, the empirical equations of Cole and Cole,[1] and Fuoss and Kirkwood [2] often give an adequate fit to the experimental behaviour. In many cases the behaviour is not symmetrical about the logarithm of the frequency of maximum loss, and the Davidson-Cole empirical relation [3, 4] gives a good fit to the experimental data. The details of the " skewed arc " representation have been reviewed by Davidson.[5] In the course of our studies on the primary (α) relaxation of amorphous polymers the non-symmetrical behaviour has become evident, and does not appear to be adequately fitted to a Davidson-Cole distribution. It is the aim of the present paper to examine a simple empirical form for the dielectric decay function which leads to non-symmetrical behaviour of the type observed in polymers.

THEORY

The complex dielectric constant ε^* is related to the normalized decay function $\gamma(t)$ by the equation [6]

$$\frac{\varepsilon^* - \varepsilon_\infty}{\varepsilon_0 - \varepsilon_\infty} = \mathscr{L}\left[-\frac{\mathrm{d}\gamma(t)}{\mathrm{d}t}\right]. \tag{1}$$

$\mathscr{L}$ indicates the Laplace transform,† $[-\mathrm{d}\gamma(t)/\mathrm{d}t]$ is the normalized transient decay current obtained when a steady electric field is removed from a sample. ε_0 and ε_∞ are the limiting low and high frequency dielectric constants of the given medium. If the decay function $\gamma(t)$ is given by the relation $\gamma(t) = \exp(-t/\tau)$, where τ is a macroscopic time constant, then eqn (1) yields the single relaxation time expression

$$(\varepsilon^* - \varepsilon_\infty)/(\varepsilon_0 - \varepsilon_\infty) = (1 + \mathrm{j}\omega\tau)^{-1}.$$

† $\mathscr{L}[F(t)] = \int_0^\infty \exp(-j\omega t)F(t)\mathrm{d}t.$

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We consider the empirical decay function

$$\gamma(t) = \exp -(t/\tau_0)^{\beta}, \tag{2}$$

$0<\beta\leqslant 1$, and τ_0 is some effective relaxation time. Insertion of (2) into (1) gives

$$\frac{\varepsilon^*-\varepsilon_\infty}{\varepsilon_0-\varepsilon_\infty} = \mathscr{L}\left[\frac{\beta}{\tau_0}\left(\frac{t}{\tau_0}\right)^{-(1-\beta)} \exp -\left(\frac{t}{\tau_0}\right)^{\beta}\right]. \tag{3}$$

The evaluation of eqn. (3) by analytical means appears complicated. However, if we choose the particular value $\beta = 0.50$, the transform is readily evaluated. For this special case, eqn (3) becomes

$$\frac{\varepsilon^*-\varepsilon_\infty}{\varepsilon_0-\varepsilon_\infty} = \tfrac{1}{2}\left(\frac{\pi}{\tau_0}\right)^{\frac{1}{2}} \mathscr{L}\left[\frac{1}{(\pi t)^{\frac{1}{2}}}\cdot \exp -2k(t)^{\frac{1}{2}}\right] \tag{4}$$

where $2k = (\tau_0)^{-\frac{1}{2}}$. The Laplace transform is now in a standard form,[7] giving

$$\frac{\varepsilon^*-\varepsilon_\infty}{\varepsilon_0-\varepsilon_\infty} = \tfrac{1}{2}\left(\frac{\pi}{\tau_0}\right)^{\frac{1}{2}} \cdot \frac{1}{(j\omega)^{\frac{1}{2}}} \exp \frac{k^2}{j\omega} \cdot \operatorname{erfc} \frac{k}{(j\omega)^{\frac{1}{2}}}. \tag{5}$$

Since $(j)^{\frac{1}{2}} = (1/\sqrt{2})(1+j)$, eqn (5) becomes

$$\frac{\varepsilon^*-\varepsilon_\infty}{\varepsilon_0-\varepsilon_\infty} = (\pi)^{\frac{1}{2}}\frac{(1-j)}{\rho} w(z). \tag{6}$$

$\rho = (8\omega\tau_0)^{\frac{1}{2}}$; $z = (1+j)/\rho$ and $w(z) = \exp(-z^2)\cdot \operatorname{erfc}(-jz)$. The values of $w(z)$ have been tabulated (ref. (7), p. 325), so $(\varepsilon^*-\varepsilon_0)/(\varepsilon_0-\varepsilon_\infty)$ may be determined as a function of $\omega\tau_0$ for $\beta = 0.50$.

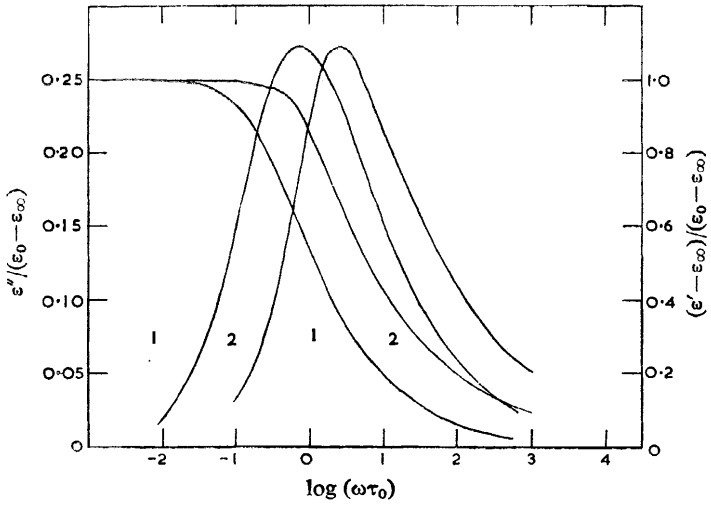

FIG. 1.—$(\varepsilon'-\varepsilon_\infty)/(\varepsilon_0-\varepsilon_\infty)$ and $\varepsilon''/(\varepsilon_0-\varepsilon_\infty)$ against log $(\omega\tau_0)$ for the present function with $\beta = 0.50$ (curves 1), and Davidson-Cole function with $\sigma = 0.322$ (curves 2).

DISCUSSION

Fig. 1 shows the plots of $(\varepsilon'-\varepsilon_\infty)/(\varepsilon_0-\varepsilon_\infty)$ and $\varepsilon''/(\varepsilon_0-\varepsilon_\infty)$ against log $(\omega\tau_0)$, for $\beta = 0.50$, as determined by eqn (6). Also shown in fig. 1 are the corresponding plots for the Davidson-Cole empirical relation with the parameter ($\sigma = 0.322$)

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chosen to give the same $\varepsilon''_{max}/(\varepsilon_0-\varepsilon_\infty)$ as the present representation with $\beta = 0.50$. These plots have the same general appearance, but differ significantly in detail, as is shown by table 1. The τ_0 values are different for the two functions, and so are their shapes, for although the half-widths of the loss factor curves are similar, their assymmetries, as measured by Δ (low)/Δ (high), are quite different. This comparison is extended to fig. 2, where the Argand plot shows a difference in shape, especially at high frequencies.

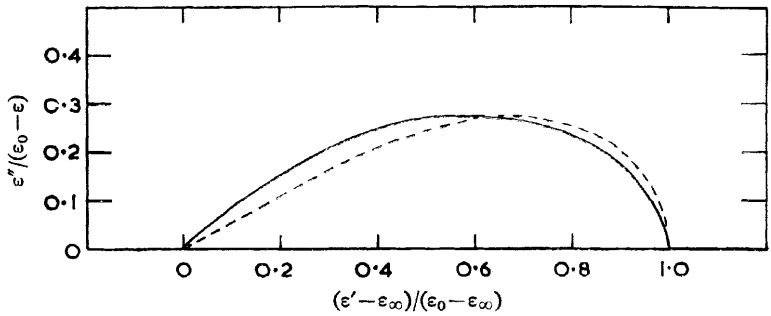

FIG. 2.—$\varepsilon''/(\varepsilon_0-\varepsilon_\infty)$ against $(\varepsilon'-\varepsilon_\infty)/(\varepsilon_0-\varepsilon_\infty)$ for the present function with $\beta = 0.50$ (solid line), and Davidson-Cole function with $\sigma = 0.322$ (dashed line).

In order to compare the function with experimental data it is desirable to have values for the function for which $\beta \neq 0.50$. The case where $\beta = 0.50$ has been evaluated exactly, and the case where $\beta = 1$ is also known, as this corresponds to the single relaxation time function. Using this information a short extrapolation has given the relaxation curves for $\beta \neq 0.5$.

TABLE 1.—A COMPARISON OF THE DAVIDSON-COLE FUNCTION AND THE PRESENT FUNCTION (EQN (6)) FOR $\varepsilon''_{max}/(\varepsilon_0-\varepsilon_\infty) = 0.272$

	$\log(\omega_{max}\tau_0)$	total half width (decade)	Δ(low)	Δ(high)	$\Delta_{(low)}/\Delta_{(high)}$
Davidson-Cole	0.40	2.02	0.70	1.32	0.53
present function	−0.15	2.12	0.90	1.22	0.74

Δ(low) and Δ(high) correspond to $|\log(\omega_i/\omega_{max})|$, where ω_i are the angular frequencies corresponding to $(\varepsilon''/\varepsilon''_{max}) = 0.50$.

The dielectric relaxation in amorphous polyethyl acrylate (PEA) has been studied in the frequency range 10^{-2}-10^5 Hz using a General Radio 1615-A bridge and a Scheiber bridge.[8] The measurements were performed at 3 pressures at 10.3°C using techniques similar to those described earlier.[9] Fig. 3 shows the experimental data plotted in normalized form, for which $\varepsilon''_{max}/(\varepsilon_0-\varepsilon_\infty) = 0.21$. Accordingly a value of $\beta = 0.38$ has been chosen for the present function, for which $\varepsilon''_{max}/(\varepsilon_0-\varepsilon_\infty)$ is also 0.21. Similarly, a value of $\sigma = 0.217$ has been chosen for the Davidson-Cole function, so that $\varepsilon''_{max}/(\varepsilon_0-\varepsilon_\infty) = 0.21$. All the information has been plotted in normalized form, and the agreement between the experimental data and the present function is reasonable, whereas the Davidson-Cole curve is more skewed. If an attempt is made to fit the α process for polypropylene oxide,[10] polyethylene terephthalate [11] and polyvinylacetate,[12] a reasonable fit is also obtained, and therefore the function may be of use for polymers.

Our main conclusion is that if $\gamma(t) = \exp -(t/\tau_0)^\beta$ is an approximately correct decay function, then it would be desirable to construct a molecular theory, giving a

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non-exponential decay of this form. A distribution of relaxation times has *not* been introduced, though this decay function would correspond to a particular distribution.

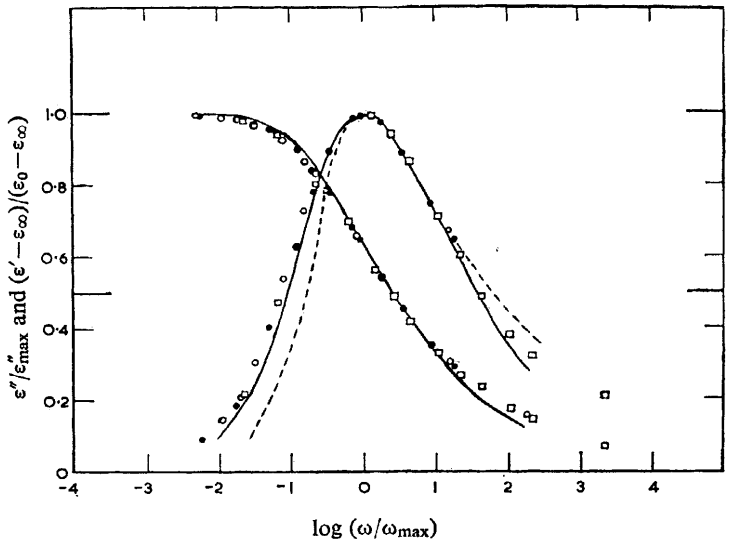

FIG. 3.—$\varepsilon''/\varepsilon''_{max}$ against log (ω/ω_{max}) for (i) present function with $\beta = 0.38$ (solid line); (ii) Davidson-Cole function with $\sigma = 0.217$ (dashed line); (iii) PEA at different applied pressures at 10.3°C. ○, 1.01×10^5 N m^{-2} (1 atm); ●, 4.20×10^7 N m^{-2} (4.16×10^2 atm); □, 8.27×10^7 N m^{-2} (8.20×10^2 atm).

We may compare the forms of $\phi(t) = -\mathrm{d}\gamma(t)/\mathrm{d}t$ for the present function, the Davidson-Cole function [13] and the Cole-Cole function.[14] These are given by eqn (7*a*), (7*b*) and (7*c*) respectively:

$$\phi(t) = \frac{\beta}{\tau_0}\left(\frac{t}{\tau_0}\right)^{-(1-\beta)} \exp-\left(\frac{t}{\tau_0}\right)^{\beta}, \tag{7a}$$

$$\phi(t) = \frac{1}{\tau\Gamma(\sigma)}\left(\frac{t}{\tau}\right)^{-(1-\sigma)} \exp\left(-\frac{t}{\tau}\right), \tag{7b}$$

$$\left.\begin{aligned}\phi(t) &= \frac{n}{\tau\Gamma(1+n)}\left(\frac{t}{\tau}\right)^{-(1-n)} &&: \text{ for } \left(\frac{t}{\tau}\right)\ll 1\\ &= \frac{n}{\tau\Gamma(1-n)}\left(\frac{t}{\tau}\right)^{-(1+n)} &&; \text{ for } \left(\frac{t}{\tau}\right)\gg 1\end{aligned}\right\} \tag{7c}$$

For short times ($t/\tau \ll 1$) which correspond to high frequencies, these three functions have the same dependence upon time, for chosen parameters, apart from a small numerical difference. However, as we proceed from the high frequency side of the loss peak, down to the low frequency side, the Davidson-Cole function falls away much more rapidly than does the Cole-Cole function, due to the exponential factor in $\phi(t)$. Our present function falls away less rapidly than does the Davidson-Cole function and the Cole-Cole function, in the low frequency region. This is because $\exp -(t/\tau_0)^\beta$ is a slower function of time than $\exp(-t/\tau)$, provided that $0<\beta\leqslant 1$.

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Finally, since we know $\gamma(t)$ and $\varepsilon^*(\omega)$ for $\beta = 0.50$, we may investigate the accuracy of the Hamon approximation,[15], i.e.,

$$\varepsilon''(\omega) = \phi(t)\omega^{-1}, \tag{8a}$$

$$\omega t = 0.1 \times 2\pi. \tag{8b}$$

These equations have been used extensively [16, 17] for transforming data obtained as a function of time, to the corresponding frequency dependent data.

When $\beta = 0.50$, then from eqn (7*a*),

$$\phi(t) = \frac{1}{2(t\tau_0)^{\frac{1}{2}}} \exp -\left(\frac{t}{\tau_0}\right)^{\frac{1}{2}}. \tag{9}$$

Substituting eqn (9) into (8*a*) gives (10*a*); and (10*b*) is equivalent to (8*b*):

$$\varepsilon''(\omega) = (1/1.256)\alpha \exp(-\alpha), \tag{10a}$$

$$\omega\tau_0 = 0.1 \times 2\pi/\alpha^2, \tag{10b}$$

where $\alpha = (t/\tau_0)^{\frac{1}{2}}$.

The loss factor curve has been evaluated using eqn (10), in the following manner. Particular values of (t/τ_0) were chosen, and the corresponding values of ε'' and $\omega\tau_0$ were then determined using (10*a*) and (10*b*) respectively. The result is compared with the true values, calculated by taking the imaginary part of eqn (6), and this is presented as the plot of ε''(Hamon)/ε''(true) in fig. 4. The agreement for log $(\omega/\omega_{max}) \geqslant 0.5$ is good, but for log $(\omega/\omega_{max}) < 0.5$ there is a discrepancy which rises to a maximum of 10 % at about log $(\omega/\omega_{max}) = -0.5$. Below this frequency, the ratio ε''(Hamon)/ε''(true) decreases to about 0.7 at log $(\omega/\omega_{max}) = -1.4$.

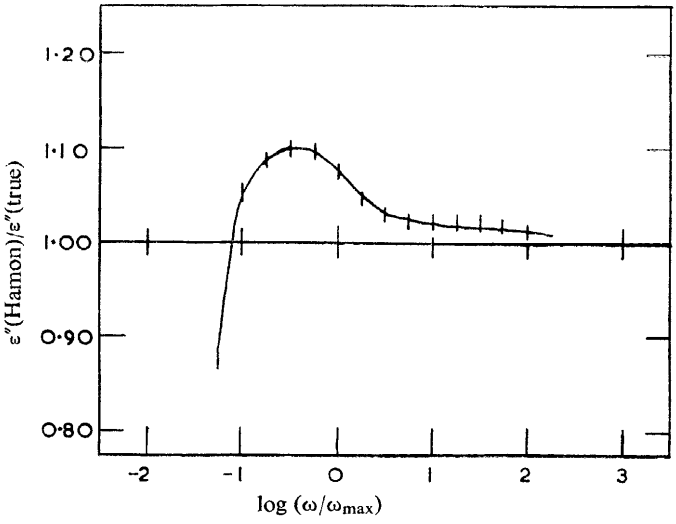

FIG. 4.—ε''(Hamon)/(ε''(true) against log (ω/ω_{max}) for the present function with $\beta = 0.50$.

A similar pattern of behaviour has been observed [18] when the Hamon approximation has been applied to the Cole-Cole distribution function. For $0.3 \leqslant n \leqslant 0.5$ the Cole-Cole, ε''(Hamon)/ε''(true) behaves in a similar manner to our function for $(\omega\tau_0) > 1$, i.e., the high frequency region, while at low frequencies $(\omega\tau_0 < 1)$ the ratio departs from unity as does our function (fig. 4). Thus, we conclude that the Hamon approximation may be used with good accuracy in the region $\omega\tau_0 > 1$, for experimental

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results which conform to our function (eqn. 7(*a*)) with $\beta = 0.5$. The approximation may be used with good accuracy, if the correction is applied from fig. 4, for results in the region $-0.5 < \log(\omega\tau_0) < 0$, but cannot be applied to transient data which would transform to loss data below $\log(\omega\tau_0) = -0.5$.

CONCLUSIONS

The empirical decay function $\gamma(t) = \exp -(t/\tau_0)^\beta$ may be transformed analytically into the complex dielectric constant if we choose $\beta = 0.5$. The resulting dispersion and absorption curves are non-symmetrical about the logarithm of the frequency of maximum loss and the behaviour lies between that obtained from the Cole-Cole and the Davidson-Cole empirical relations. The present decay function with $\beta = 0.38$ gives a reasonably satisfactory representation of the α relaxation in polyethyl acrylate at 10.3°C, and it is suggested that the α relaxation in other amorphous polymers could be fitted to the present function for suitably chosen distribution parameters. The accuracy of the Hamon approximation has been investigated for $\beta = 0.5$ and is applicable to the frequency range $\log(\omega\tau_0) > -0.5$, but fails at lower frequencies.

The authors thank Imperial Chemical Industries Ltd. for a grant for the purchase of dielectric measuring equipment, and the Science Research Council for the award of a studentship to D. C. W.

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